

Deep-fake Forensics TSYP11 CS - YP Challenge

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Abstract—TSYP 11, a Tunisian congress, annually unveils a challenge for the Computer Science chapter within the IEEE Tunisia section. In its 11th edition, the challenge centers on replicating and refining the outcomes outlined in a research paper titled "Shallow- and Deep-Fake Image Manipulation Localization Using Deep Learning" authored by Junbin Zhang, Hamidreza Tohidypour, Yixiao Wang, and Panos Nasiopoulos. The Goal of the challenge is to reproduce the results of the paper, improve them, and make it sustainable. Our work exist in the github Repository : <https://github.com/HadjSassi/ShallowDeepFakesLocalization>

Index Terms—IEEE CS Chapter, TSYP11, CS Challenge, Deep Fake Forensics, UPerNet, Deep learning

INTRODUCTION

The paper titled "Shallow- and Deep-Fake Image Manipulation Localization Using Deep Learning"[1] represents a pioneering effort in developing a model capable of simultaneously detecting both shallow and deep fakes. Notably, it introduces a novel approach by incorporating the UperNet[2] model as a foundation and enhancing it with an additional treatment called BayerConv to ensure versatility in various detection scenarios.

The comparative analysis within the paper evaluates the developed model against the UperNet[2] in terms of deepfake detection and localization. Additionally, it assesses its performance against the MVSS-Net[3] model specifically designed for shallow fake detection.

In our paper, we will exclusively focus on deep fake forensics, disregarding shallow fake considerations. This decision aligns with the TSYP challenge parameters, which exclusively emphasize deep fake classification.

I. PROJECT OVERVIEW

In outlining the project, we begin by introducing the specific challenges we address. Subsequently, we discuss the solution presented in the research paper titled "Shallow- and Deep-Fake Image Manipulation Localization Using Deep Learning." [1] We then detail our process of filtering and selecting pertinent components, focusing on the aspects integral to our project. The subsequent

section elaborates on how we faithfully replicated the results demonstrated in the III part of the research paper. Following this, we explore various optimization attempts aimed at improving and refining the outcomes. Lastly, we elucidate on the potential implementation of this project as a startup idea, examining its viability and application in real-world scenario.

II. PROBLEMATIC

The proliferation of deepfake technology poses a critical challenge in the realm of image forensics. With the increasing sophistication of deep fake algorithms, the ability to discern authentic from manipulated content has become exceedingly difficult. This project addresses the pressing need for robust and accurate deep fake detection methods, aiming to develop a solution that can effectively identify and localize manipulated content in images. The challenge lies in creating a model that not only replicates the results achieved by state-of-the-art deepfake detection systems but also explores avenues for optimization and enhancement to stay ahead of evolving deceptive techniques.

III. PROPOSED SOLUTION BY THE PAPER "SHALLOW- AND DEEP-FAKE IMAGE MANIPULATION LOCALIZATION USING DEEP LEARNING"

The proposed solution involves designing a novel deep learning network for detecting and localizing manipulated areas in both shallow- and deep-fake images. The foundation of this solution is based on the UPerNet[2], a state-of-the-art image segmentation method. To enhance its capabilities, BayerConv is integrated into the network, allowing for the extraction and utilization of features from both the original images and the noise distribution. This modification ensures a comprehensive approach to decision-making in image localization. The performance evaluation demonstrates the effectiveness of the proposed method, achieving 82% accuracy in real/fake categorization and 75% accuracy in localization for both shallow- and deepfake test images. The solution outperforms existing state-of-the-art methods retrained on the provided dataset, showcasing its potential for robust shallow- and deep-fake detection in both image and video content.

In Figure 1, the network architecture is illustrated, showcasing the key components and modifications introduced to the original UPerNet[2] for effective image manipulation localization. The figure outlines the sequence of operations performed by the network. The initial step involves utilizing BayerConv, a set of learnable high-pass filters, to extract the noise distribution of the input images.

Subsequently, two ResNet50 backbones are employed to extract multi-level feature representations. These feature maps from each level are concatenated layer-wise to capture comprehensive information. Four convolutional layers are then applied to these concatenated feature maps for improved fusion. The highest-level information is fed into a Pyramid Pooling Module (PPM) head for the detection task, which focuses on overall image-level information. Simultaneously, the other levels of features are integrated into the Feature Pyramid Network (FPN), a process similar to the original UPerNet[2]. At the highest resolution, a localization head is attached to generate masks highlighting manipulated areas. The green rectangles in the figure denote the modifications introduced to the original UPerNet[2], particularly the incorporation of BayerConv, which plays a crucial role in facilitating features from both the original images and the noise distribution, contributing to the network’s comprehensive decision-making process in manipulation localization.

IV. PROJECT FILTRATION AND SELECTION

During this critical phase, our primary focus is on meticulous filtering and the strategic selection of components extracted from the research paper to precisely align with our project objectives. To kickstart this process, we obtained the relevant dataset exclusively dedicated to deepfakes. The dataset was sourced from the project’s GitHub readme file, accessible at this link 2.

Subsequently, we executed a refinement of the dataset by updating paths, deliberately preserving only those entries concluding with "df" to denote instances of deepfakes. Notably, we consciously excluded entries with "sf" for shallow fake and any other hybrid types, underscoring our project’s specific emphasis on deepfake detection within its defined scope. File kept are :

- paths_test_df.txt
- paths_train_df.txt
- paths_val_df.txt

V. REGENERATION RESULTS

The primary objective of the challenge is to replicate the research paper’s project titled "Shallow- and Deep-Fake Image Manipulation Localization Using Deep Learning"[1] and achieve results consistent with those presented in the original research. However, it’s important to note that our focus is solely on deep learning forensics, which imposes a limitation to the test set results for deepfakes within the Upernet[2] with BayerConv project. This targeted approach ensures a

concentrated effort on the specific aspects relevant to our project’s goals, optimizing our resources and efforts for the deepfake detection framework.

In the upcoming table I, we will showcase the outcomes obtained from our testing of the project. The results are identical to the 8th trained model of the project, named 8.pth.

	Test Set DeepFakes			
UPerNet with BayerConv	F1			AUC
	Pixel-Level	Image-Level	Combined	
	0.92	0.95	0.94	0.99

TABLE I: Results of Upernet with BayerConv on Deep-Fake Test Set

VI. OPTIMIZATION ATTEMPTS

A. Dataset Extension

Our quest for an enriched dataset led us to a comprehensive exploration of the web, with a specific emphasis on Kaggle 3—an eminent hub for diverse and high-quality data science projects. By leveraging the resources available, we successfully expanded our existing dataset by an additional 2GB. This augmentation included the integration of 5492 fake testing frames and 5413 real testing frames, broadening the spectrum of scenarios for evaluation.

To fortify the foundation of our deep learning model, we made substantial additions to the training set. Now comprising 70,000 fake training frames and an equal number of real training frames, this enlarged dataset aims to capture a more comprehensive range of features and variations inherent in deepfake and authentic content.

Equally important, the validation set underwent a significant enhancement. With 19,600 fake validation frames and 19,800 real validation frames, our model will undergo rigorous testing to ensure robust performance across various conditions.

This meticulous expansion strategy not only bolsters the quantity of our dataset but also ensures a diversity of data points, ultimately contributing to the improved training, testing, and validation of our deep learning model for effective deepfake detection.

B. Brute Force On params α and β

In our pursuit of refining the performance of the deep learning model, an exhaustive exploration of parameter optimization was undertaken, specifically targeting the α , β , and epoch values. This comprehensive search involved a series of brute-force attempts to uncover the most effective combination of parameters. However, it was observed that on the majority of these trials, the model’s

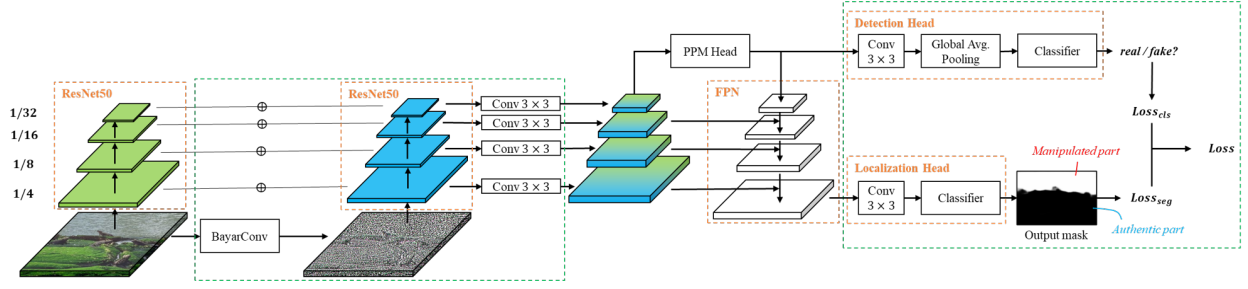


Fig. 1: Structure of our network. "⊕" means we concatenate the feature maps at each layer of the two ResNet50. Each concatenated feature map is then fed into a different convolutional layer to fuse the features. Components enclosed in green rectangles are our modifications on top of the original UPerNet.

Type:	File folder
Location:	C:\Users\mahdi\Desktop\IEEE\TSYP\CS\ShallowDeep
Size:	13.6 GB (14,682,221,607 bytes)
Size on disk:	13.7 GB (14,742,831,104 bytes)
Contains:	30,459 Files, 21 Folders

Fig. 2: Dataset for the Deep Fake

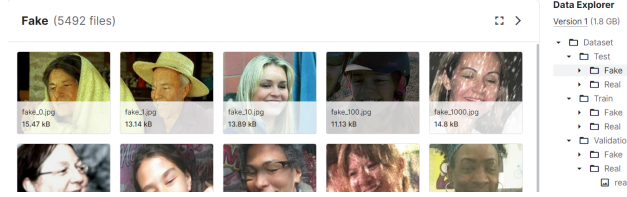


Fig. 3: Dataset expansion with deepfake and real images

performance deteriorated. Amidst these challenges, a breakthrough emerged when adjustments were made to the α and β values, setting them at 0.03 and 0.19, respectively. Remarkably, this singular alteration led to a notable enhancement, with the F1 metric registering a substantial improvement. Notably, the area under the curve (AUC) surged to an impressive 98%, as detailed in the forthcoming table, signifying a pivotal moment in the optimization process. However, it is worth noting that the AUC of the previous model was 99%, adding a note of melancholy to this achievement.

In the upcoming table II, we will showcase the best outcomes obtained from our testing of the project after the alterations we made.

	Test Set DeepFakes			
UPerNet with BayerConv	F1			AUC
	Pixel-Level	Image-Level	Combined	
	0.93	0.95	0.95	

TABLE II: Results of Upernet with BayerConv on Deep-Fake Test Set

VII. IMPLEMENTATION INTO A STARTUP IDEA

The third goal of the TSYP11 CS Challenge is to explore and implement a startup idea, along with creating its Business Model Canvas (BMC)⁴. Our envisioned venture involves the development of a web application dedicated to detecting the authenticity of video content, targeting social media platforms such as

- Facebook
- LinkedIn
- ...

Additionally, media companies, including entities like

- Mozaïque FM
- El Hiwar El Tounsi
- ...

stand to benefit from our solution. Legal professionals, the lawyers, recognizing its value as evidentiary support, are identified as a crucial clientele.

For casual users, we propose free access to basic services, but premium features, such as localizing fake segments in a video and appending accurate captions, will be available through a paid subscription, priced at, for instance, 1 Tunisian Dinar per minute of processing. To foster customer relations, we offer free service after a specified number of uses, ensuring a balance between accessibility and monetization.

Our commitment to ongoing customer satisfaction involves an After Sales Service that continuously updates and enhances the application based on user feedback. Casual customers will be offered a free trial to familiarize themselves with the application before deciding on a subscription.

To realize this ambitious project, we anticipate financial allocations for skilled human resources, technological infrastructure (including server hosting), and marketing efforts. The latter will involve showcasing trending videos on which our algorithms have been applied, collected through our key partnerships with social media platforms. The bulk of our expenditure will be directed towards application development, hosting, maintenance, and marketing, ensuring the success and sustainability of our innovative solution.

CONCLUSION

In conclusion, we are excited about the prospects of our envisioned web application for detecting video authenticity, a venture that not only addresses the challenge of deepfake proliferation across social media and media companies but also aligns seamlessly with the goals of the TSYPI1 CS Challenge. We have successfully achieved the challenge objectives by comprehensively understanding the scientific paper 'Shallow- and Deep-Fake Image Manipulation Localization Using Deep Learning.' [1] We regenerated the results presented in the paper, explored optimization strategies, and ultimately derived valuable insights that propelled us towards the creation of a robust startup idea. Our strategic business model, catering to diverse user segments and underpinned by a commitment to continuous improvement, positions us for success in the dynamic landscape of digital content integrity. As we embark on this innovative journey, we are not only meeting the TSYPI1 CS Challenge goals but also contributing meaningfully to the advancement of deepfake detection technologies.

REFERENCES

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Business Model Canvas

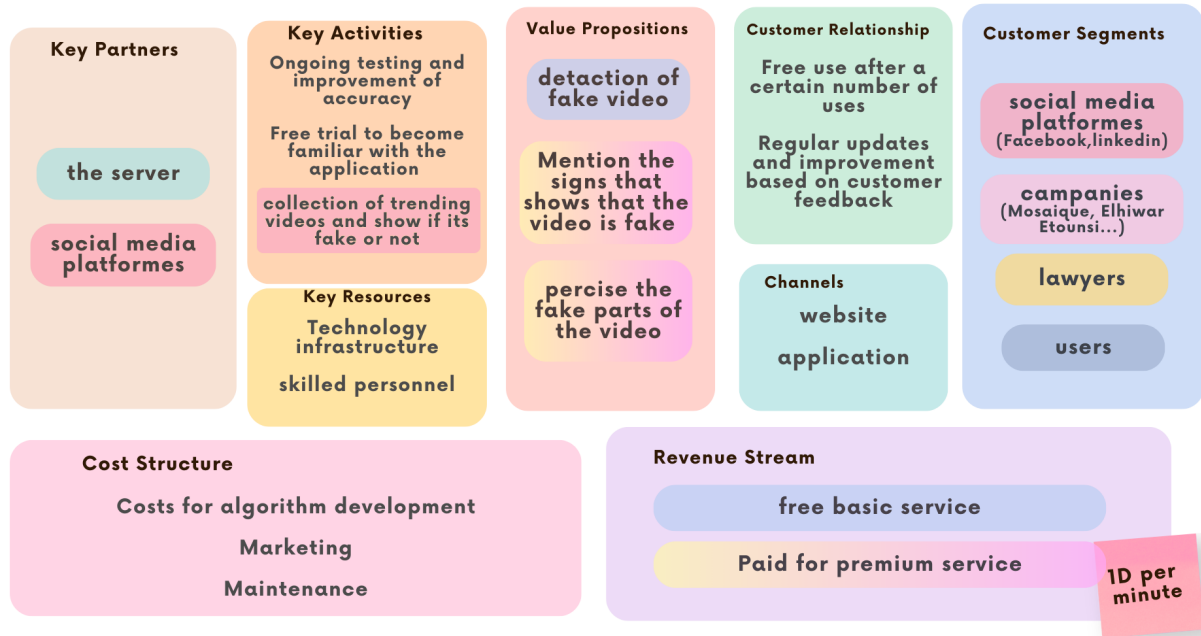


Fig. 4: Business Model Canvas